

## NIVALA, Finland

WMO: 029050

Lat: 63.917N

Lon: 24.967E

Elev: 266

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 90-00

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -15.0      | -8.6 | -19.4                         | 1.9 | -14.9 | -12.7 | 2.8 | -8.4 | 26.9                  | 37.4 | 22.9 | 36.7 | 2.8                   | 160  | 0.667 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.5                   | 77.7            | 63.4 | 74.2 | 61.6 | 71.2 | 60.1 | 65.2                | 73.1 | 63.4 | 71.5 | 61.7 | 69.0 | 7.1                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 62.4                            | 85.0 | 69.3 | 60.2 | 78.5 | 66.3 | 58.4 | 73.5 | 64.5 | 30.2          | 72.5 | 28.9 | 72.0 | 27.7 | 68.9 | 72.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 20.5          | 17.7 | 15.4 | -21.7                      | 82.7 | 6.2                | 3.4 | -26.2                                              | 85.1 | -29.9      | 87.1 | -33.4      | 89.0 | -37.9      | 91.4 |     |     |
| (5)               |      |      | -22.0                      | 68.7 | 6.1                | 2.5 | -26.4                                              | 70.5 | -30.0      | 72.0 | -33.5      | 73.4 | -37.9      | 75.3 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 37.1   | 18.4  | 17.4  | 25.8  | 34.7  | 45.0  | 56.0  | 59.9  | 56.3  | 46.3  | 36.5  | 26.0  | 21.7  |  |  |
|                                                                        | DBStd       | 17.24  | 11.62 | 13.56 | 8.54  | 8.07  | 7.50  | 6.62  | 5.06  | 5.85  | 6.62  | 8.39  | 9.53  | 11.60 |  |  |
|                                                                        | HDD50       | 5497   | 979   | 912   | 749   | 462   | 197   | 20    | 1     | 12    | 150   | 418   | 720   | 877   |  |  |
|                                                                        | HDD65       | 10200  | 1444  | 1332  | 1214  | 910   | 621   | 278   | 172   | 275   | 560   | 882   | 1170  | 1342  |  |  |
|                                                                        | CDD50       | 800    | 0     | 0     | 0     | 2     | 42    | 200   | 308   | 207   | 40    | 1     | 0     | 0     |  |  |
|                                                                        | CDD65       | 28     | 0     | 0     | 0     | 0     | 0     | 9     | 14    | 5     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 307    | 0     | 0     | 0     | 0     | 12    | 102   | 137   | 56    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 35     | 0     | 0     | 0     | 0     | 0     | 11    | 22    | 2     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 6.7    | 7.2   | 7.0   | 7.3   | 6.8   | 6.7   | 6.6   | 6.4   | 5.9   | 6.3   | 7.3   | 6.6   | 6.7   |  |  |
|                                                                        | PrecAvg     | 23.7   | 1.6   | 1.3   | 1.2   | 1.3   | 1.8   | 2.5   | 3.1   | 2.8   | 2.1   | 2.2   | 2.0   | 1.9   |  |  |
|                                                                        | PrecMax     | 30.4   | 3.0   | 2.5   | 2.0   | 2.8   | 3.4   | 4.5   | 5.4   | 5.6   | 4.9   | 4.2   | 3.8   | 3.9   |  |  |
| (15) Precipitation                                                     | PrecMin     | 18.0   | 0.4   | 0.2   | 0.2   | 0.3   | 0.3   | 0.8   | 0.8   | 1.1   | 0.7   | 0.5   | 0.2   | 0.7   |  |  |
|                                                                        | PrecStd     | 3.2    | 0.7   | 0.6   | 0.5   | 0.6   | 0.9   | 1.1   | 1.1   | 1.2   | 1.2   | 1.0   | 0.8   | 0.8   |  |  |
|                                                                        | DB          | 40.3   | 42.2  | 46.7  | 62.8  | 75.7  | 81.4  | 83.4  | 78.5  | 68.7  | 55.3  | 45.1  | 39.4  |       |  |  |
|                                                                        | MCWB        | 36.9   | 37.3  | 40.6  | 50.9  | 59.6  | 65.1  | 68.0  | 64.1  | 60.2  | 51.7  | 43.4  | 37.4  |       |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 36.7   | 38.6  | 42.0  | 56.7  | 70.8  | 77.1  | 77.9  | 75.3  | 63.6  | 51.8  | 42.2  | 37.6  |       |  |  |
|                                                                        | 2% MCWB     | 34.8   | 36.2  | 36.9  | 47.0  | 56.3  | 62.5  | 64.4  | 63.1  | 56.6  | 49.3  | 40.7  | 36.1  |       |  |  |
|                                                                        | 5% DB       | 34.6   | 36.0  | 39.1  | 52.1  | 65.0  | 72.9  | 74.1  | 71.9  | 59.8  | 49.6  | 39.8  | 36.1  |       |  |  |
|                                                                        | 5% MCWB     | 33.3   | 34.2  | 35.1  | 44.3  | 52.9  | 60.1  | 62.1  | 61.8  | 54.6  | 47.3  | 38.4  | 34.8  |       |  |  |
|                                                                        | 10% DB      | 32.8   | 34.2  | 36.7  | 47.2  | 59.9  | 69.2  | 70.7  | 68.3  | 57.1  | 47.4  | 37.1  | 34.5  |       |  |  |
|                                                                        | 10% MCWB    | 31.8   | 32.8  | 34.0  | 41.0  | 50.2  | 58.5  | 60.5  | 59.7  | 53.4  | 45.2  | 35.8  | 33.4  |       |  |  |
|                                                                        | 0.4% WB     | 37.2   | 39.0  | 41.5  | 52.1  | 60.0  | 68.1  | 69.8  | 66.1  | 61.0  | 52.6  | 44.1  | 37.9  |       |  |  |
|                                                                        | 0.4% MCDB   | 39.6   | 41.1  | 45.8  | 61.3  | 70.9  | 78.8  | 81.1  | 73.2  | 66.6  | 54.7  | 44.9  | 38.7  |       |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 35.2   | 36.5  | 37.7  | 47.5  | 57.2  | 64.3  | 66.2  | 64.0  | 58.1  | 49.7  | 40.8  | 36.3  |       |  |  |
|                                                                        | 2% MCDB     | 36.6   | 38.3  | 40.9  | 55.9  | 69.9  | 72.8  | 73.4  | 72.3  | 62.1  | 51.4  | 41.9  | 37.3  |       |  |  |
|                                                                        | 5% WB       | 33.6   | 34.6  | 35.9  | 44.6  | 54.1  | 61.9  | 63.7  | 62.5  | 55.8  | 47.7  | 38.6  | 35.0  |       |  |  |
|                                                                        | 5% MCDB     | 34.5   | 36.0  | 38.3  | 51.0  | 63.4  | 70.4  | 71.4  | 70.9  | 59.0  | 49.2  | 39.7  | 35.9  |       |  |  |
|                                                                        | 10% WB      | 31.9   | 33.0  | 34.4  | 41.8  | 50.8  | 59.4  | 62.1  | 60.4  | 53.3  | 45.3  | 35.8  | 33.4  |       |  |  |
|                                                                        | 10% MCDB    | 32.9   | 34.1  | 36.3  | 46.8  | 58.6  | 67.9  | 68.9  | 67.2  | 56.4  | 47.0  | 36.9  | 34.2  |       |  |  |
|                                                                        | MDBR        | 11.8   | 12.4  | 14.6  | 16.0  | 18.6  | 17.2  | 16.5  | 16.7  | 14.5  | 9.1   | 8.4   | 10.2  |       |  |  |
|                                                                        | 5% DB       | 10.0   | 10.6  | 15.2  | 23.4  | 28.1  | 23.8  | 23.6  | 23.2  | 17.7  | 10.2  | 8.7   | 8.5   |       |  |  |
| (36) Mean Daily Temperature Range                                      | 5% DB       | 9.5    | 9.6   | 11.1  | 14.7  | 15.5  | 12.1  | 11.6  | 12.2  | 11.5  | 8.2   | 7.9   | 8.1   |       |  |  |
|                                                                        | 5% WB       | 9.5    | 10.2  | 12.1  | 22.1  | 25.4  | 20.7  | 20.0  | 20.9  | 15.0  | 9.5   | 8.7   | 8.6   |       |  |  |
|                                                                        | MCWBR       | 9.3    | 9.5   | 9.2   | 14.1  | 14.6  | 11.5  | 10.8  | 11.8  | 10.5  | 8.4   | 8.2   | 8.4   |       |  |  |
|                                                                        | MCWBR       | 9.3    | 9.5   | 9.2   | 14.1  | 14.6  | 11.5  | 10.8  | 11.8  | 10.5  | 8.4   | 8.2   | 8.4   |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.218  | 0.282 | 0.310 | 0.351 | 0.346 | 0.348 | 0.369 | 0.353 | 0.329 | 0.308 | 0.361 | 0.356 |       |  |  |
|                                                                        | taud        | 1.987  | 2.136 | 2.196 | 2.317 | 2.443 | 2.457 | 2.416 | 2.456 | 2.512 | 2.378 | 2.637 | 2.631 |       |  |  |
|                                                                        | Ebn at Noon | 113    | 198   | 245   | 261   | 275   | 276   | 266   | 260   | 244   | 193   | 105   | 58    |       |  |  |
|                                                                        | Edh at Noon | 9      | 20    | 28    | 32    | 31    | 31    | 31    | 28    | 21    | 16    | 8     | 5     |       |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 48     | 223   | 692   | 1186  | 1548  | 1704  | 1589  | 1201  | 682   | 257   | 61    | 19    |       |  |  |
|                                                                        | RadStd      | 5      | 41    | 67    | 115   | 168   | 148   | 158   | 123   | 73    | 46    | 12    | 2     |       |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (43 neighbors) | +0.92 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page